

Electric Field

$E \rightarrow$ Electric field $\rightarrow$ N/C

A region around an electric charge (moving - stationary)



E lines are outward



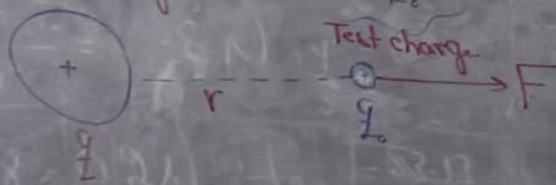
E lines are inward

The electric field at any point is defined as the electric force at that point divided by q_{test} .

$$E = \frac{F}{q_{test}}$$

Source charge

Test charge



$$F = k_e \frac{Qq}{r^2}$$

$$E = \frac{F}{q} = \frac{k_e \frac{Qq}{r^2}}{q} \Rightarrow E = k_e \frac{Q}{r^2}$$

Example 1.2

Electric Field

Find the resultant electric force on q due to q_1 and q_2

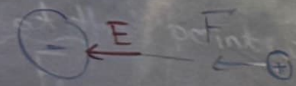
Test charge $\rightarrow$ N/C

- ① having small value $\rightarrow$ to do not affect the electric field to be measured
- ② Positive

The electric field at any point is defined as the electric force that a unit positive charge would experience if placed at that point.

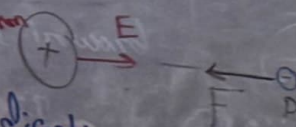
When a test charge is positive:

E and F are in the same direction



When a test charge is negative:

E and F are in opposite direction
+180



Electric Field

Find the resultant electric force on q due to q_1 and q_2

• electric field is a property of a source charge, that means the test charge is not necessary to exist.

• A test charge serves as a detector of the electric field.

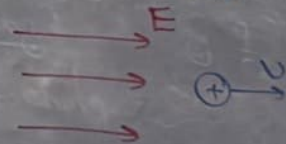
E lines properties:

- for positive charge: E lines $\rightarrow$ out
for negative charge: E lines $\rightarrow$ in

- number of E lines depends on a magnitude of a source charge
direction of E lines depends on a sign of a source charge
- No two E lines can cross each other



Straight-parallel lines



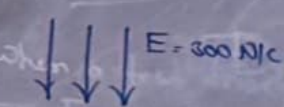
Electric Field

Example 1.10

0.6 g ball carries a charge

It is suspended by a thread in $E = 300 \text{ N/C}$

is directed down. What is the tension in the thread if a charge is



①

$$q = 8 \mu\text{C}$$

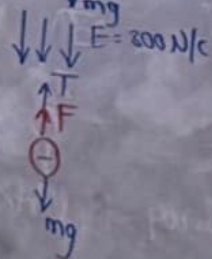
$$T = mg + F$$

$$T = 5.9 \times 10^{-3} + 2.4 \times 10^{-3} = 8.3 \times 10^{-3} \text{ N} = 8.3 \text{ mN}$$

$$\therefore E = \frac{F}{q} \rightarrow F = qE = 8 \times 10^{-6} \times 300$$

$$F = 2.4 \times 10^{-3} \text{ N}$$

$$mg = 0.6 \times 10^{-3} \times 9.8 = 5.9 \times 10^{-3} \text{ N}$$



②

$$q = -8 \mu\text{C}$$

$$T + F = mg$$

$$T = mg - F = 5.9 \times 10^{-3} - 2.4 \times 10^{-3} = 3.5 \times 10^{-3} \text{ N} = 3.5 \text{ mN}$$

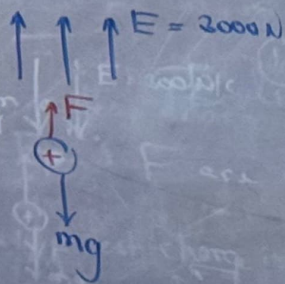
Example 1.5

Electric Field

Example 1.5

0.2 g ball hangs from a thread in $E = 3 \text{ kN/C}$ directed upward

What is the charge if a tension in the thread is



(a)

$$T = 0$$

$$T + F = mg$$

$$F = mg$$

$$qE = mg$$

(b)

$$T = 4 \text{ mN}$$

$$q = \frac{mg}{E} = \frac{0.2 \times 10^{-3} \times 9.8}{3000} = 0.65 \times 10^{-6} \text{ C} \rightarrow 0.65 \mu\text{C}$$

$$mg = 0.2 \times 10^{-3} \times 9.8 = 1.96 \times 10^{-3} \text{ N}$$

$$T = mg + F$$

$$T = mg + qE \rightarrow q = \frac{T - mg}{E} = \frac{4 \times 10^{-3} - 1.96 \times 10^{-3}}{3000} = 0.68 \times 10^{-6} \text{ C} \rightarrow q = -0.68 \mu\text{C}$$

